Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr.S60_Nu-XL-500 & ALL | JEE-MAIN | Date: 01-01-2026

Time: 7.30AM to 10.30AM | IGTM-04 | Max. Marks:300

KEY SHEET

MATHEMATICS

1.	3	**2.**	3	**3.**	3	**4.**	2	**5.**	4
6.	2	**7.**	1	**8.**	2	**9.**	3	**10.**	3
11.	3	**12.**	1	**13.**	4	**14.**	1	**15.**	1
16.	3	**17.**	2	**18.**	1	**19.**	2	**20.**	3
21.	34	**22.**	34	**23.**	216	**24.**	2035	**25.**	16

PHYSICS

26.	3	**27.**	1	**28.**	2	**29.**	4	**30.**	4
31.	3	**32.**	2	**33.**	1	**34.**	1	**35.**	2
36.	3	**37.**	2	**38.**	1	**39.**	4	**40.**	2
41.	1	**42.**	4	**43.**	1	**44.**	4	**45.**	1
46.	1	**47.**	1	**48.**	1200	**49.**	1	**50.**	2

CHEMISTRY

51.	1	**52.**	1	**53.**	1	**54.**	2	**55.**	2
56.	4	**57.**	4	**58.**	1	**59.**	1	**60.**	4
61.	4	**62.**	1	**63.**	3	**64.**	4	**65.**	1
66.	2	**67.**	4	**68.**	1	**69.**	1	**70.**	4
71.	372	**72.**	2435	**73.**	74	**74.**	6	**75.**	1

SOLUTIONS
MATHEMATICS

1. Centroid of vertices (1,3), (3,1), (2,4) is $\left(\frac{1+3+2}{3},\frac{3+1+4}{3}\right)=\left(2,\frac{8}{3}\right)$

Image of (2,8/3) wrt the line $x+2y=2$ is (α,β)

$$\frac{h-2}{1}=\frac{k-\frac{8}{3}}{2}=\frac{-2\left(2+2\left(\frac{8}{3}\right)-2\right)}{4+1}\Rightarrow\frac{h-2}{1}=\frac{k-\frac{8}{3}}{2}=\frac{-32}{15}$$

$$h-2=-\frac{32}{15}\Rightarrow h=\frac{-2}{15}\quad\text{and}\quad\frac{k-\frac{8}{3}}{2}=-\frac{32}{15}\Rightarrow k=-\frac{24}{15}$$

$$\therefore 15(\alpha-\beta)=15\left[\left(\frac{-2}{15}\right)-\left(\frac{-24}{15}\right)\right]=22$$

2. $\underbrace{A,B,C,D,\ldots\ldots\ldots\ldots,M}_{\substack{12\text{ Letters}\\ \text{We should select 2.}}}\ \underbrace{N,O,P,\ldots\ldots\ldots\ldots,X}_{\substack{13\text{ Letters}\\ \text{We should select 2.}}},Y,Z.$

$$\underbrace{\therefore {}^{12}C_2\times1\ldots\ldots\ldots\ldots M\ldots\ldots\ldots\ldots{}^{13}C_2\times1}_{\text{Arrangement in Alphabetical Order}}$$

$$The\,ways={}^{12}C_2\times{}^{13}C_2=\frac{12(11)}{2}\times\frac{(13)12}{2}=5148$$

3. $Sec^{-1}x=a\in[0,\pi]-\left\{\frac{\pi}{2}\right\}$ $\qquad\cos ec^{-1}x=\frac{\pi}{2}-a$

$$16\left[\left(\sec^{-1}x\right)^2+\left(\operatorname{cosec}^{-1}x\right)^2\right]=16\left[a^2+\left(\frac{\pi}{2}-a\right)^2\right]=16\left[2a^2-\pi a+\frac{\pi^2}{4}\right]$$

$$\max]_{a=\pi}=16\left[2\pi^2-\pi^2+\frac{\pi^2}{4}\right]=20\pi^2\qquad\min]_{a=\pi/4}=16\left[\frac{2\pi^2}{16}-\frac{\pi^2}{4}+\frac{\pi^2}{4}\right]=2\pi^2$$

$$\therefore Sum=22\pi^2$$

4. $f(x)=7\tan^8x+7\tan^6x-3\tan^4x-3\tan^2x$

$$f(x)=7(\tan^8x+\tan^6x)-3(\tan^4x+\tan^2x)=7(\tan^6x\sec^2x)-3(\tan^2x\sec^2x)$$

$$I_1=\int_0^{\pi/4}f(x)dx=7\int_0^{\pi/4}\tan^6x\sec^2x.dx-3\int_0^{\pi/4}\tan^2x\sec^2x.dx$$

$$I_1=7\left[\frac{\tan^7x}{7}\right]_0^{\pi/4}-3\left[\frac{\tan^3x}{3}\right]_0^{\pi/4}=7\left(\frac{1}{7}-0\right)-3\left(\frac{1}{3}-0\right)=0$$

$$I_2=\left[x\left(\tan^7x-\tan^3x\right)\right]_0^{\pi/4}-\int_0^{\pi/4}\left(\tan^7x-\tan^3x\right)dx$$

$$I_2=0-\int_0^{\pi/4}\tan^3x\left(\tan^2x-1\right)\left(\tan^2x+1\right)dx$$

Let $\tan x=t\Rightarrow\sec^2x.dx=dt$

$$I_2=-\int_0^1t^3\left(t^2-1\right)dt=-\left[\frac{t^6}{6}-\frac{t^4}{4}\right]_0^1=\frac{1}{12}\Rightarrow12I_2=1\qquad\therefore7I_1+12I_2=0+1=1$$

5. $f(x)=p^x \Rightarrow f'(x)=p^x \log p$

$f'(0)=p^0.\log p=4a \Rightarrow p=e^{4a}$

$f(x)=e^{4ax}$

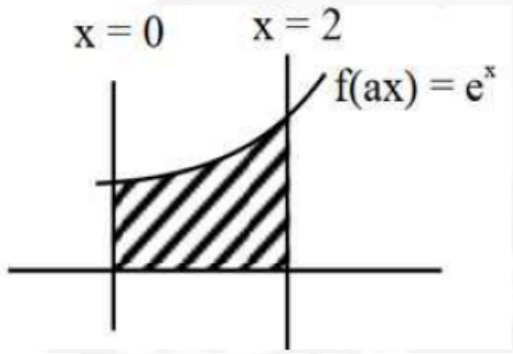
$f''(x)-3af'(x)-f(x)=0$

$16a^2e^{4ax}-12a^2e^{4ax}-e^{4ax}=0 \Rightarrow 4a^2-1=0 \Rightarrow a=\frac{1}{2}$

$f(x)=e^{2x}$

$0 \le y \le f(1/2)=e^x$.

$R.A=\int_0^2 e^x.dx=e^2-e^0=e^2-1$



6. $m=1,\ n=2,3,.....,10 \rightarrow 9$

$m=2,\ n=3,5,7,9 \rightarrow 4$

$m=3,\ n=4,5,7,8,10 \rightarrow 5$

$m=4,\ n=5,7,9 \rightarrow 3$

$m=5,\ n=6,7,8,9 \rightarrow 4$

$m=6,\ n=7 \rightarrow 1$

$m=7,\ n=8,9,10 \rightarrow 3$

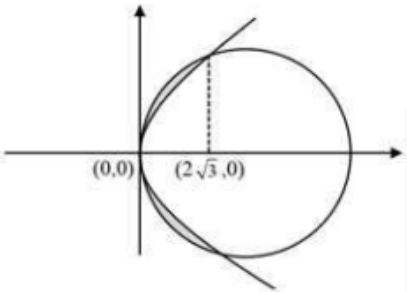
$m=8,\ n=9 \rightarrow 1$

$m=9,\ n=10 \rightarrow 1$

31

7. $\text{Area}=2\left[\frac{1}{4}\pi\left(2\sqrt{3}\right)^2-\left(\int_0^{2\sqrt{3}}\sqrt{2\sqrt{3}}\times\sqrt{x}.dx\right)\right]$

$=2\left[3\pi-\sqrt{2\sqrt{3}}\left[\frac{x^{3/2}}{3/2}\right]_0^{2\sqrt{3}}\right]=2\left[3\pi-\frac{2\sqrt{2\sqrt{3}}\left(2\sqrt{3}\right)^{3/2}}{3}\right] \qquad =2\left[3\pi-\frac{3\times 2^3}{3}\right]=6\pi-16$



8. $f(x+y)=f(x)f'(y)+f(y)f'(x)$.....(1)

Put $x=0, y=0$

$f(0)=2f(0)f'(0)\Rightarrow f'(0)=\frac{1}{2}$

Put $y=0$ in equation(1)

$f(x)=f(x)f'(0)+f(0)f'(x)$

$f(x)=\frac{f(x)}{2}+f'(x)\Rightarrow\frac{f(x)}{2}=f'(x)$

$\Rightarrow\frac{1}{2}=\frac{1}{f(x)}f'(x)\Rightarrow\int\frac{1}{2}dx=\int\frac{1}{f(x)}f'(x).dx$

$\Rightarrow\log_e f(x)=\frac{x}{2}+c$

Where $f(0)=1\Rightarrow 0=0+c\Rightarrow c=0$

$\therefore\sum_{r=1}^{100}\log_e f(n)=\sum_{r=1}^{100}\left(\frac{n}{2}\right)=\frac{1}{2}\left(\frac{100(100+1)}{2}\right)=2525$

9. Let equation had two solution be x_1 and x_2. $e^{5(\log_e x)^2+3}=x^8$

Taking log on both sides, we get

$5(\log_e x)^2-8\log_e x+3=0$ $\Rightarrow\log_e x_1x_2=\frac{8}{5}\Rightarrow x_1x_2=e^{8/5}$

10. $y^2dx+\left(x-\frac{1}{y}\right)dy=0\Rightarrow\frac{dx}{dy}+x\left(\frac{1}{y^2}\right)=\frac{1}{y^3}$

$IF=e^{\int\frac{1}{y^2}dy}=e^{\frac{1}{y}-1}$

$xe^{\frac{-1}{y}}=\int e^{\frac{-1}{y}}\left(\frac{1}{y}\right)\left(\frac{1}{y^2}dy\right)=-\int e^z(z).dz=-\left[ze^z-e^z\right]+c$

$\Rightarrow xe^{\frac{-1}{y}}=-e^z[z-1]+c=-e^{\frac{-1}{y}}\left[\frac{-1}{y}-1\right]+c$

$\Rightarrow x=\left(\frac{1}{y}+1\right)+ce^{\frac{1}{y}}$ $\because x(1)=1\Rightarrow 1+2+ce\Rightarrow c=\frac{-1}{e}$

$x=\left(\frac{1}{y}+1\right)+e^{\frac{1}{y}-1}$

$y=\frac{1}{2}\Rightarrow\frac{1}{y}=2$ $x=(2+1)-e^{2-1}=3-e$

11. $2a=\left|\sqrt{(1-1)^2+(14-6)^2}-\sqrt{(1-1)^2+(-12-6)^2}\right|\Rightarrow 2a=|8-18|\Rightarrow a=5$

$2ae=\sqrt{(1-1)^2+(14-(-12))^2}=26\Rightarrow e=\frac{26}{10}=\frac{\sqrt{a^2+b^2}}{5}$

$\sqrt{25+b^2}=13\Rightarrow b=12$

length of latus rectum $=\frac{2b^2}{a}=\frac{2(144)}{5}=\frac{288}{5}$

12. $|z|=1,\ \arg(z_1)=\frac{-\pi}{4},\ \ \arg(z_1)=\frac{-\pi}{4},\ \ \arg(z_3)=\frac{\pi}{4}$

$$\left|z_1\overline{z_2}+z_2\overline{z_3}+z_3\overline{z_1}\right|^2=\left|\left(\frac{1}{\sqrt{2}}-\frac{i}{\sqrt{2}}\right)+\left(\frac{1}{\sqrt{2}}-\frac{i}{\sqrt{2}}\right)+\left(\frac{1}{\sqrt{2}}-\frac{i}{\sqrt{2}}\right)^2\right|^2$$

$$=\left|2\left(\frac{1}{\sqrt{2}}-\frac{i}{\sqrt{2}}\right)++\left(0+\frac{2i}{2}\right)\right|^2=\left|\sqrt{2}-i\sqrt{2}+i\right|^2=\left|\sqrt{2}+i\left(1-\sqrt{2}\right)\right|^2$$

$$=2+\left(1-\sqrt{2}\right)^2=5-2\sqrt{2}=\alpha+\beta\sqrt{2}$$

$$\therefore \alpha^2+\beta^2=(5)^2+(-2)^2=29$$

13. $r>1$

$$a_1a_5=28\Rightarrow a_1\left(a_1r^4\right)=28\Rightarrow a_1^2r^4=28\Rightarrow a_1r^2=2\sqrt{7}$$

$$a_2+a_4=29\Rightarrow a_1r+a_1r^3=29\Rightarrow a_1r(1+r^2)=29$$

$$a_1r=a_2=28,\ a_2=1=a_1r$$

$$a_6=a_1r^5=\frac{1}{\sqrt{28}}\left(\sqrt{28}\right)^5=784$$

14.

HHH	$H\underline{HT}$	$\underline{HT}H$	THH	$\underline{HTT}$	$T\underline{HT}$	TTH	TTT
0	1	1	0	1	1	0	0
$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$

$X=x$	0	1
$P(X=x)$	4/8	4/8

$$\mu=\sum_i x_iP(X=x_i)=\frac{4}{8}=\frac{1}{2}$$

$$\sigma^2=\sum_i\left(x_i^2P(X=x_i)\right)-\left(\mu^2\right)=0+\frac{1}{8}+0+\frac{1}{8}+\frac{1}{8}-\frac{1}{4}=\frac{1}{4}$$

$$\therefore 64\left(\mu+\sigma^2\right)=64\left(\frac{1}{2}+\frac{1}{4}\right)=48$$

15. by Baye's Theorem

$$P\left(\frac{\text{1st is black}}{\text{2nd is black}}\right)=\frac{\frac{6}{10}\times\frac{5}{9}}{\left(\frac{4}{10}\times\frac{6}{9}\right)+\left(\frac{6}{10}\times\frac{5}{9}\right)}=\frac{5}{9}=\frac{m}{n}$$

$$\therefore m+n=5+9=14$$

16. Centre is $C_1(-2, 2)$

Radius $r_1 = 2$

$C_2(2, 5)$ $r_2 = r$

$|r_1 - r_2| < C_1C_2 < r_1 + r_2$

$|r-2| < 5 < r+2$

$r+2 > 5 \Rightarrow r > 3$ or $|r-2| < 5 \Rightarrow r < 7$

$3 < r < 7$

$(\alpha, \beta) = (3, 7)$

$\therefore 3\beta - 2\alpha = 21 - 6 = 15$

17. $P = (2t+1, 3t+2, 4t+3)$ $Q = (3s+2, 4s+4, 5s+5)$

D.R's of PQ $= (3s-2t+1, 4s-3t+2, 5s-4t+2)$

D.R's of $L_1 = (2,3,4)$

D.R's of $L_2 = (3,4,5)$

Then, D.R's of PQ=(1,-2,1)

$$\frac{3s-2t+1}{1} = \frac{4s-3t+2}{-2} = \frac{5s-4t+2}{1}$$

On solving, $s = \frac{-1}{6}, t = \frac{1}{3}$

$P = \left(\frac{5}{3}, 3, \frac{13}{3}\right)$ $Q = \left(\frac{3}{2}, \frac{10}{3}, \frac{25}{6}\right)$

Equation of PQ is $\frac{x - \frac{5}{3}}{1} = \frac{y-3}{-2} = \frac{z - \frac{13}{3}}{1}$

$\left(\frac{14}{3}, -3, \frac{22}{3}\right)$ lies on PQ.

18. $$T_n = S_n - S_{n-1} = \frac{(2n-1)(2n+1)(2n+3)(2n+5)}{64} - \frac{(2n-3)(2n-1)(2n+1)(2n+3)}{64}$$

$$= \frac{8(2n-1)(2n+1)(2n+3)}{64} = \frac{(2n-1)(2n+1)(2n+3)}{8}$$

$$\frac{1}{T_r} = \frac{8}{(2r-1)(2r+1)(2r+3)}$$

$$\lim_{n\to\infty} \sum_{r=1}^{n} \left(\frac{1}{T_r}\right) = \lim_{n\to\infty} \sum_{r=1}^{n} \left(\frac{8}{(2r-1)(2r+1)(2r+3)}\right)$$

$$= 2 \lim_{n\to\infty} \sum_{r=1}^{n} \left(\frac{1}{(2r-1)(2r+1)} - \frac{1}{(2r+1)(2r+3)}\right) = 2 \lim_{n\to\infty} \left(\frac{1}{(1)(3)} - \frac{1}{(2n+1)(2n+3)}\right) = \frac{2}{3}$$

19. $y = x^2 + px - 3$

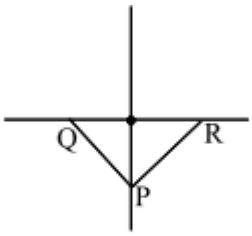
$\Rightarrow x = 0 \Rightarrow y = -3 \quad P(0,-3)$

Centre C=(-1, -1) and radius $r = CP = \sqrt{5}$

Equation of circle is $(x+1)^2 + (y+1)^2 = 5$

$y = 0 \Rightarrow x^2 + 2x - 3 = 0 \Rightarrow x = -3, 1$

Q(-3, 0) and R(1,0)



$\Delta = \frac{1}{2}(QR)(OP) = 6$

20. No. of equivalence relations =Bell no.

Bell no.

For n=1	1		
For n=2	1	2	
For n=3	2	3	5

No. of equivalence relations=5

21. f is continuous and differentiable at x=1

$-3a - 2 = a^2 + b$ and $-6a = b$

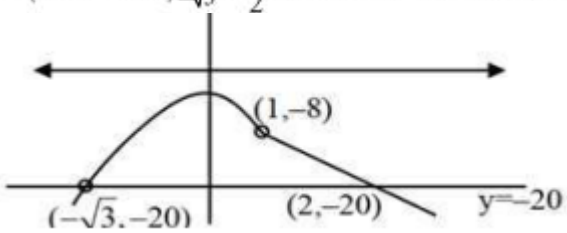
$\Rightarrow a = 1$ or 2, $\because a > 1, a = 2, b = -12$

$a^2 - 3a + 2 = 0 \qquad \Rightarrow a = 1$ or 2, $\because a > 1, a = 2, b = -12$

$$\therefore f(x) = \begin{cases} -6x^2 - 2, x < 1 \\ 4 - 12x, x \geq 1 \end{cases}$$

$$R.A. = \int_{-\sqrt{3}}^{1} \left[\left(-6x^2 - 2\right) - (-20)\right] dx + \left(\text{Area of } \Delta ABC\right)$$

$$= \left(-2x^3 + 18x\right)_{-\sqrt{3}}^{1} + \frac{1}{2}(1)(12) = 22 + 12\sqrt{3} = \alpha + \beta\sqrt{3} \qquad \therefore \alpha + \beta = 22 + 12 = 34$$



22. $|A| = -2$

$$\left|3adj\left(-6adj\left(3A\right)\right)\right| = 3^3\left|adj\left(-6adj\left(3A\right)\right)\right|$$

$$= 3^3\left|-6Adj(3A)\right|^2 = 3^3(-6)^6\left|Adj(3A)\right|^2$$

$$= 3^3.2^6.3^6\left|3A\right|^4 = 3^9.2^6.3^{12}\left|A\right|^4 = 2^{10}3^{21} = 2^{m+n}.3^{mn}$$

$m+n=10$, $mn=21 \Rightarrow m=7, n=3$

$4m+2n = 28+6 = 34$

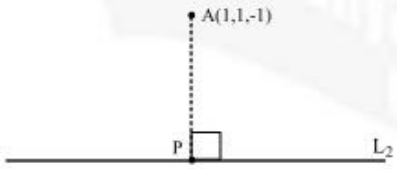
23. let $L_1 : \frac{x-1}{3} = \frac{y-1}{-1} = \frac{z+1}{0} = t$ and $L_2 : \frac{x-2}{2} = \frac{y}{0} = \frac{z+4}{\alpha} = s$

$B = (3t+1, -t+1, -1) = (2s+2, 0, \alpha s-4)$

$3t+1 = 2s+2,$ $\quad -t+1 = 0 \Rightarrow t = 1,$ $\quad -1 = \alpha s - 4$

$\Rightarrow 4 = 2s+2 \Rightarrow s = 1$ $\quad \Rightarrow \alpha = 3$

$\therefore B = (4, 0, -1)$



Let $P = (2\lambda+2, 0, 3\lambda-4)$

D.R.s of AP$=(2\lambda+1, -1, 3\lambda-3)$

$$AP \perp L_2 \Rightarrow 2(2\lambda+1)+0+3(3\lambda-3) = 0 \Rightarrow \lambda = \frac{7}{13}$$

$$P = \left(\frac{40}{13}, 0, \frac{-31}{13}\right)$$

$$PB^2 = \frac{468}{169}$$

$$\therefore 26\alpha PB^2 = 26(3)\left(\frac{468}{169}\right) = 216$$

24. ${}^{n+1}C_{r+1} = \frac{n+1}{r+1}\,{}^{n}C_r$

Putting $n=11$ and r by $2r+1$

$${}^{11}C_{2r+1}.\frac{12}{2r+2} = {}^{12}C_{2r+2} \Rightarrow \frac{{}^{11}C_{2r+1}}{2r+2} = \frac{{}^{12}C_{2r+2}}{12}$$

$$\therefore \sum_{r=0}^{5}\frac{{}^{11}C_{2r+1}}{2r+2} = \frac{1}{12}\sum_{r=0}^{5}{}^{12}C_{2r+2} = \frac{1}{12}\left[{}^{12}C_2 + {}^{12}C_4 + \ldots\ldots + {}^{12}C_{12}\right]$$

$$= \frac{1}{12}\left[\left({}^{12}C_0 + {}^{12}C_2 + {}^{12}C_4 + \ldots\ldots + {}^{12}C_{12}\right) - {}^{12}C_0\right] = \frac{2^{11}-1}{12} = \frac{m}{n}$$

$$\therefore m-n = (2^{11}-1)-12 = 2035$$

25.

$$\vec{c}=\left(\frac{\vec{b}.\vec{a}}{|\vec{a}|^2}\right)\vec{a}=\vec{c}=\left(\frac{\lambda+8}{9}\right)\vec{a}$$

$$|\vec{a}+\vec{c}|=7\Rightarrow\left|\vec{a}+\left(\frac{\lambda+8}{9}\right)\vec{a}\right|=7\Rightarrow|\lambda+17|=21\Rightarrow\lambda=4(\lambda>0)\qquad\vec{c}=\frac{4}{3}\vec{a}$$

Area of parallelogram $=|\vec{b}\times\vec{c}|=\frac{4}{3}|\vec{b}\times\vec{a}|=\frac{4}{3}\sqrt{144}=16$

PHYSICS

26.

$$W=\vec{F}.\vec{S}=0$$

$$\left(2\hat{i}+b\hat{j}+\hat{k}\right).\left(\hat{i}-2\hat{j}-\hat{k}\right)=0$$

$$2-2b-1=0$$

$$b=1/2$$

27.

$$T=2\pi\sqrt{\frac{L}{g}}\text{ and }g=\frac{GM}{R^2}$$

$$\frac{g_1}{g_2}=\frac{M_1}{M_2}\times\left(\frac{R_2}{R_1}\right)^2$$

$$=\frac{\cancel{M}}{4\cancel{M}}\times\left(\frac{2\cancel{R}}{\cancel{R}}\right)^2=\frac{1}{\cancel{4}}\times\frac{\cancel{4}}{1}=1$$

$$g_1=g_2$$

$\therefore$ Time period depends on length of pendulum and acceleration due to gravity (g) and it is independent of mass of pendulum .

28.

$$\bar{\tau}=\bar{r}\times\bar{F}\qquad\bar{F}=2i+j+2k$$

$$\bar{r}=i+j+k\qquad\tau=\begin{vmatrix}i & -j & k\\1 & 1 & 1\\2 & 1 & 2\end{vmatrix}$$

$$=i(2-1)-j(2-2)+k(1-2)$$

$$=\hat{i}-o-\hat{k}\qquad=\hat{i}-\hat{k}$$

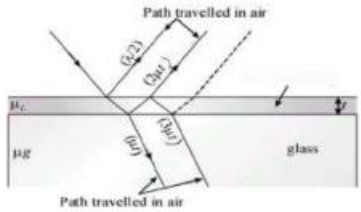
29.

$$\gamma=1+\frac{2}{f}\qquad\underline{for\ Diatomic}\ f=5\qquad\gamma_1=1+\frac{2}{5}=\frac{7}{5}=1.4\rightarrow(1)$$

If vibrational model are taken into account , f=7

$$\gamma_2=1+\frac{2}{7}=\frac{9}{7}=1.28\qquad\gamma_2=1.3\rightarrow(2)\qquad\therefore\ \gamma_2<\gamma_1$$

30.



Path difference for transmission $\Delta = 2\mu t$

For max transmission $2\mu t = n\lambda \ (n = 1,2,3....)$

$t = \frac{\lambda}{2\mu}$ $t_{\min} = \frac{550nm}{2\times 2} = \frac{550}{4}$ $t_{\min} = 137.5nm$

31. Extensive variable depends on the size or mass if the system

Ex : Volume , Total, mass, extropy, internal energy neat capacity .

Intensive variables do not depends on the size or mass of the system.

Ex : Temperature pressure, specific heat capacity density etc .

32. Capacity (C)= $\frac{q}{V} = \frac{q}{W/q} = \frac{q^2}{W}$ $= \frac{C^2}{ML^2T^{-2}} = \left[C^2M^{-1}L^{-2}T^2\right]$

33. A^+point of projection

$KE = \frac{1}{2}mv^2$ $= \frac{1}{\cancel{2}} \times 100 \times 10^{-3} \times \cancel{20} \times 20$

$KE = 20J \rightarrow (1)$ $\frac{at\ highest\ point}{u =_u x = u\cos 60^0}$ $= \frac{20}{2} = 10ms^{-1}$

$KE = \frac{1}{2} \times 100 \times 10^{-3} \times 100$ $= \frac{10}{2} = 5J$

$\Delta KE = 20 - 5 = 15J$

34. $\overrightarrow{E_0} = \frac{1}{4\pi\varepsilon_0}\frac{2p}{r^3} \rightarrow (1)$ $E = \frac{1}{4\pi\varepsilon_0}\frac{p}{(2r)^3}$

$E = \frac{1}{4\pi\varepsilon_0}\frac{p}{8r^3} \times \frac{2}{2}$ $E = \frac{1}{4\pi\varepsilon_0}\frac{2p}{r^3} \times \frac{1}{16}$

$E_B = \frac{E_0}{16} \rightarrow (2)$ $\frac{Potentaial}{v = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2}}$

$at\ B\ , \theta = 90^0$ $v_B = 0$

35. $\beta = \frac{\lambda D}{d}$ $\therefore \beta \alpha \lambda$

36. $mvr = \frac{nh}{2\pi}$ $\frac{mv^2}{r} = qvB$

$r = \frac{mv}{qb}$ $r = \frac{nh}{2\pi r}qb$ $r^2 = \frac{nh}{2\pi qB}$

For first excited state =n=2 $r = \sqrt{\frac{\cancel{2}h}{\cancel{2}\pi eB}}$ $r = \sqrt{\frac{h}{\pi eB}}$

37. $I_m = \frac{v_m}{\sqrt{R^2 + (X_L - X_C)^2}}$

At Resonance $X_L = X_C$ $I_m = \frac{v_m}{R} = I_0$ $I_m = \frac{v_m}{2R} = \frac{I_0}{2}$

38. $A_1V_1 = A_2V_2$ $\pi(2)^2(2) = \pi(1)^2 V_2$ $8 = v_2$

39. $[P] = \frac{[MLT^{-2}]}{[L^2]} = [ML^{-1}T^{-2}]$

$[D] = [ML^{-3}]$

$[S] = \frac{[MLT^{-2}]}{[L]} = [MT^{-2}]$

$[M^0L^0T^1] = [ML^{-1}T^{-2}]^a[ML^{-3}]^b\ [MT^{-2}]^c$

$a + b + c = 0$ .....(i)

$-a - 3b = 0$ $\Rightarrow a = -3b$ .....(ii)

$-2a - 2c = 1$ $\Rightarrow a + c = -\frac{1}{2}$ .....(iii)

From (i) and (iii) $b = \frac{1}{2}$

From (ii) $a = -\frac{3}{2}$

$a = -\frac{3}{2}, b = \frac{1}{2}, c = 1$

40. $P = \frac{1}{f} = 4D$

$P^1 = \frac{1}{2f}$ $= \frac{4}{2}D$ $= 2D$

41. $\rho = \frac{m}{v} = \frac{m}{\pi r^2 l}$ $\frac{\Delta\rho}{\rho} = \frac{\Delta m}{m} + 2\left(\frac{\Delta r}{r}\right) + \frac{\Delta l}{l}$

$= \frac{0.003}{0.60} + 2\left(\frac{0.01}{0.5}\right) + \left(\frac{0.05}{10}\right) = 0.005 + 0.04 + 0.005$ $= 0.05$ $= 5\%$

42. $\varepsilon = -\frac{d\phi}{dt}$ $\therefore \phi = BA$

$\varepsilon = -B\left(\frac{dA}{dt}\right)$

$\frac{dA}{dt} = cons\tan t$ and $B = cons\tan t$ $\therefore \varepsilon$ is also constant .

43. $E = K_m + \phi$

$E = 2eV + 1eV$

$\Rightarrow E = 3eV$

$E\alpha\frac{1}{\lambda}$

$\frac{E_2}{E} = \frac{\lambda_1}{\lambda_2} = \frac{\lambda}{\frac{\lambda}{2}} = 2, E_2 = 2E = 6ev$

$E_2 - 2E = 6eV$

$KE_2 = E_2 - \phi = 6 - 1 = 5eV$

44. For position $= \sqrt{v^2 - 2gl(1-\cos\theta)}$

If $\theta = 30^0$ $v_B = \sqrt{100 - 2(10)(2)(1-\cos 30^0)}$

$= \sqrt{100 - 40\left(1-\sqrt{3/2}\right)}$ $= \sqrt{100 - 20\left(2-\sqrt{3}\right)}$

$v_B^2 = 100 - 40 + 20\sqrt{3}$ $v_B^2 = 60 + 20\sqrt{3}$

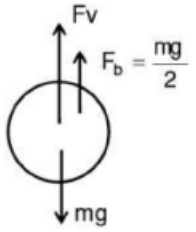
For position ‘ C’

$v_c = \sqrt{100 - 2(10)(2)(1-\cos 120^0)}$ $= \sqrt{100 - 40\left(1+\frac{1}{2}\right)}$

$= \sqrt{100 - 40(3/2)}$ $= \sqrt{100 - 60}$ $= \sqrt{40}$

$v_c^2 = 40$ $\frac{KE_B}{KE_C} = \frac{v_B^2}{v_c^2} = \frac{60 + 20\sqrt{3}}{40}$ $= \frac{3+\sqrt{3}}{2}$

45.

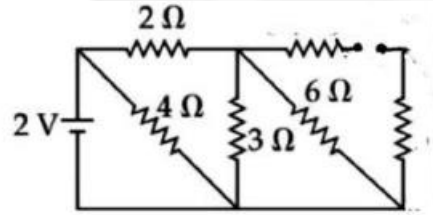


$f_V + f_B = Mg$

$f_V = Mg - F_B \qquad = Mg - \rho_L(V)g$

$= Mg - \frac{\rho}{2}vg \qquad = Mg - \frac{Mg}{2} \qquad = \frac{Mg}{2}$

46.



$\frac{3\times 6}{3+6} = \frac{18}{9} = 2\Omega$

$2+2 = 4\Omega \qquad \frac{4\times 4}{4+4} = \frac{16}{8} = 2\Omega$

current $I = \frac{\cancel{2}}{\cancel{2}} = 1A$

47.

$B_x = \frac{\mu_0 i}{2\pi r} \qquad = \frac{4\pi\times 10^{-7}\times 5}{2\pi\times 10\times 10^{-2}}$

$B_x = 10^{-5} \rightarrow (1)$ inward

$B_y = \frac{\mu_0 i}{2\pi r} = \frac{4\pi\times 10^{-7}\times 4}{2\pi\times 4\times 10^{-2}} \qquad B_y = 2\times 10^{-5} \rightarrow (2)$ outward

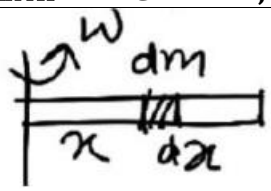
$B_{net} = B_y - B_x \qquad = 2\times 10^{-5} - 10^{-5} \qquad B_{net} = 1\times 10^{-5}T \qquad x = 1$

48.

$I_d = \varepsilon_0 A\frac{dE}{dt} \qquad = \varepsilon_a A_0 \frac{d}{dt}\left(\frac{q}{A\varepsilon_0}\right) \qquad I_D = \frac{A_0}{A}\left(\frac{dq}{dt}\right)$

$I_D = \frac{A_0}{A}I_C \qquad I_D = \frac{3.2}{16}(6) \qquad = 0.2(6) \qquad = 1.2$ Amp $=1200$ mA.

49.



$$\int_0^F dF = \int_0^L (dm)x\omega^2 \qquad dm = \rho A(dx)x\omega^2$$

After integration we can write

$$F = 2m\omega^2 \frac{l}{2}\{l=1\} \quad \Rightarrow F = m\omega^2 \qquad \Rightarrow \omega = \sqrt{\frac{F}{m}} \text{ So } \alpha = 1$$

50. $r = \frac{mv}{qB}$

$$B = \frac{mV}{qr} = \frac{1.67\times10^{-27}\times2\times10^{5}}{1.6\times10^{-19}\times2\times10^{-2}}$$

$$B \Rightarrow 0.1T$$

$$E = VB = 2\times10^{5}\times0.1$$

$$E = 2\times10^{4}\,V/m = \alpha\times10^{4}\,V/m \Rightarrow \alpha = 2$$

CHEMISTRY

51. After mixing the two solutions in equal volumes the conc of the salts ion reduced to half.

Opt-1 : $\left[1\times10^{-2}\right]\times\left[1\times10^{-2}\right] = 10^{-4}M$

$10^{-4} > 5.2\times10^{-7}$

Ionic product $> Ksp$

Hence ppt is formed.

Opt-2 $\left[\frac{3.6\times10^{-3}}{2}\right]\left[\frac{5\times10^{-4}}{2}\right]$

$1.8\times10^{-3}\times2.5\times10^{-4}$

$=1.8\times2.5\times10^{-7}$

$=4.5\times10^{-7} < Ksp$

Hence ppt is not formed.

Opt-3 $\left[\frac{1.5\times10^{-4}}{2}\right]\left[\frac{1.5\times10^{-3}}{2}\right]$

$=0.75\times0.75\times10^{-8}$

$=0.5625\times10^{-8} < Ksp$ hence ppt is not formed.

Opt-4 $\left[\frac{2.0\times10^{-4}}{2}\right]\left[\frac{0.8\times10^{-3}}{2}\right]$

$=1\times10^{-4}\times0.4\times10^{-3}$

$0.4\times10^{-7} < Ksp$

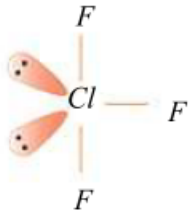
$\therefore$ ppt is not formed.

52. $CH_3-\overset{*}{C}H(Br)-CH_2-CH_3 \rightarrow CH_3-CH=CH-CH_3 \xrightarrow{Br_2}$

(A) (B)

53. Vanillin reacts with $NaOH$ and undergoes acid base reaction with phenolic hydroxyl group.
Present in it forming vanillin phenoxide which is more soluble in water. Vanillin also reacts with Tollens reagent due the presence of aldehydic group.
Vanillin does not undergoes self aldol condensation due to absence of a $\alpha-$ hydrogen.

54. NF_3 : No. of lone pairs =1
SO_2 : No. of lone pairs =1
NH_3 : No. of lone pairs=1
SF_4 : No. of lone pairs =1
XeF_2 : No. of lone pairs =3
ClF_3 : No. of lone pairs =2
XeF_2 is planar molecule & net dipole moment is zero.
ClF_3



it has T shape molecule no. of σ bonds =3
No. of lone pairs =2
Hybridization = sp^3d

55. A) $2Cu^{+2}+\left[Fe(CN)_6\right]^{-4} \rightarrow Cu_2\left[Fe(CN)\right]$ Copper ferro cyanide
(Chocolate brown colour ppt)
B) $4Fe^{+3}+3\left[Fe(CN)_6\right]^{-4} \rightarrow Fe_4\left[Fe(CN)_6\right]_3$ Ferric Ferro cyanide
(Prussian blue ppt)
C) $2Zn^{+2}+\left[Fe(CN)_6\right]^{4-} \rightarrow Zn_2\left[Fe(CN)_6\right]$ Zinc ferro cyanide
(white ppt)
D) $MgCl_2$ does not react with $K_4\left[Fe(CN)_6\right]$ to form a blue ppt. magnesium ferrocyanide is usually white ppt
E) $Ba^{+2}+\left[Fe(CN)_6\right]^{-4} \rightarrow Ba_2\left[Fe(CN)_6\right]$ barium ferrocyanide
(white ppt).

56. Metallic radius of $Ga < Al$
Ionic radius of $Ga^{+3} > Al^{+3}$

57. For an ideal gas solution

$P_T = P_A^0 \times + P_B^0 \times B$

$A = \frac{1}{4}, \times B = \frac{3}{4}$

$500 = 200 \times \frac{1}{4} + P_B^0 \times \frac{3}{4}$

$P_B^0 = 600mm$ of Hg

and least volatile compound is A

58. % of $= \frac{12}{44} \times \frac{1.46}{1} \times 100$

= % of $c = 39.81$

% of $H = \frac{2}{18} \times \frac{0.567}{1} \times 100 = 6.3$

Remaining percentage is oxygen = 53.9

C	H	O
$\frac{39.81}{12}$	$\frac{6.3}{1}$	$\frac{53.9}{16}$
3.3	6.3	3.3
1	2	1

The empirical formula is CH_2O molecular weight is =12+2+16=30

59. $K_H = \frac{P}{X_A}$ at constant of temperature

But $K_H \alpha T$

$\frac{K_H}{P} = \frac{1}{X_A}.T$

$\frac{K_H}{P} = \frac{1}{X_A}.T^0C$

$y = mx$

60. The rate of hydrolysis of carboxylic acid derivative is in the order
acid chlorides > acid anhydride > Ester > amide.
The order is based on electrophilic nature of carbonyl group and leaving ability of the groups.

61. All naturally occurring alpha amino acids except glycine are optically active because the contains chiral carbon atoms. Where as in glycine chiral carbon is absent.
Glutamic acid and aspartic acid as both possess a carboxylic group on their side chain making them acidic amino acids.
Amino acid, cysteine can undergoes dimerization due to presence of free SH group to form disulphide bond.
During dimerization of cysteine oxidation of two SH groups results in the formation of disulphide bonds.

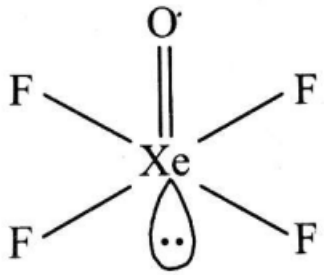
62. Fluorine has a relatively low electron affinity compared to other halogens due to its small size and high electron density (greater $e^- - e^-$ repulsions)

63. In octahedral complexes, when $\Delta_0 < P$ high spin complexes are formed.
When $\Delta_0 > P$ low spin complexes are formed.
In tetrahedral complexes because of $\Delta_t < P$, low spin complexes are rarely formed. Mostly high spin complexes are formed.

64. II Carbon is more stable due to resonance
I Carbon is least stable due to free radical present on sp carbon

65. Order of basicity of ethyl amines in aqueous solution $2^0 > 3^0 > 1^0$ which is based on inductive effect steric hindrance and solvation effect.
Between ammonia and hydrazine, NH_3 is more basic than N_2H_4 because availability of lone pairs in ammonia is more than N_2H_4 . In N_2H_4 $-I$ effect NH_2 group decreasing availability of lone pair on other nitrogen.

66. Rarest monoatomic & non-radio active element in xenon.
X: 1st electro negative =F
Y : 2nd electro negative =O
Formula : $XeOF_4$



$XeOF_4$ molecule has a square pyramidal shape.

67. Since gas is expanding from bulb 'A' to 'B' work is done by the gas
$\therefore\ dw \neq 0$

68. $CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$
No. of moles of $HCl = \frac{M \times w}{1000}$
$= \frac{0.76 \times 250}{1000}$
No. of HCl moles = 0.19 mole of HCl.
$2 \times 36.5g$ of HCl $\rightarrow 111g$ of $CaCl_2$
$0.19 \times 36.5g$ _____?
$\frac{0.19 \times 111 \times 36.5}{2 \times 36.5} = 10.545g$

69. a,c,d,e,h are aromatic because they are satisfied Huckel's rule of aromaticity
Where as b & g are anti aromatic
f is non aromatic due to non-planarity.

70. 1) Radius of 4th orbit is four times larger tan that of 2nd orbit

$\therefore$ radius ratio $= \frac{n_4^2}{n_2^2} = \frac{(4)^2}{(2)^2} = \frac{16}{4} = 4$

this statement is correct

2) Radius of 8th orbit is four times larger than that of 4th orbit

$\therefore$ radius ratio $= \frac{(n8)^2}{n_4^2} = \frac{8^2}{4^2} = \frac{64}{16} = 4$

This statement is correct

3) Radius of third orbit is nine time larger than that of 1st orbit

Radius ratio $= = \frac{(n3)^2}{n_1^2} = \frac{(3)^2}{1^2} = \frac{9}{1} = 9$

This statement is correct

4) Radius of 6th orbit is three times larger than that of 4th orbit

Radius ratio $= = \frac{(n6)^2}{n_4^2} = \frac{6^2}{4^2} = \frac{36}{16} = 2.25$

Not 3

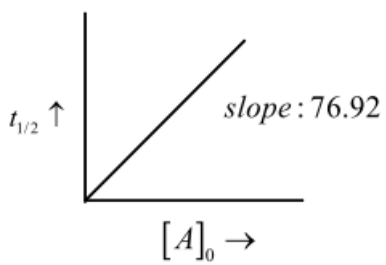
This statement is incorrect.

$\therefore$ final answer is 4.

71. 1 mole of antiviral compound (P) weighs -372gm.

so, 0.1 mol of given antiviral compound weighs $372 \times 10^{-1} g$

72.



For zero order

$t_{1/2} \alpha [A]_0$

$t_{1/2} = \frac{[A_0]}{2k_0}$

Slope (m) $= \frac{1}{2k_0}$

$\frac{1}{2k_0} = 76.92$

$k_0 = \frac{1}{76.92 \times 2}$

$k_0 = 0.0065 = 65 \times 10^{-4}$

$k_0 = \frac{[A]_0 - [A]_t}{t}$

$t = 10 \min$

$65 \times 10^{-4} = 2.5 - [A]_t$

$[A]_t = 2.5 - (65 \times 10^{-4} \times 10)$

$[A]_t = 2.5 - 65 \times 10^{-3}$

$= 2.5 - 6.5 \times 10^{-2}$

$= 2.435 = 2435 \times 10^{-3}$

73. Ideal gas is Pv=nRT

Equilibrium constant Kp is defined in terms of partial pressures.

Change in moles of $CH_3OH = +0.04mol$

Change in moles of $CO = -0.04$ moles

Change in the moles of H_2 is $-2\times 0.04 = -00.08mol$

$\therefore$ moles of CO at equilibrium $= 0.1 - 0.04 = 0.06mol$

Moles of H_2 at equilibrium initial $n_{H_2\text{initial}} - 0.08$

Moles of CH_3OH at equilibrium =0.04

$n_{total} = n_{CO} + n_{H_2} + n_{CH_3OH}$

$= 0.06 + n_{H_2\text{initial}} - 0.08 + 0.04$

$n_{total} = n_{H_2 initial} + 0.02$

$\therefore\ P_{total} v = n_{total} RT$

$5\times 2 = \left(n_{H_2\text{intial}} + 0.02\right)\times 0.08\times 500$

$10 = \left(n_{H_2\text{intial}} + 0.02\right)\times 40$

$n_{H_2\,\text{int}ial} + 0.02 = \frac{10}{40} = 0.25$

$n_{H_2\,\text{int}ial} = 0.25 - 0.02 = 0.23mol$

$n_{H_2\,equilibrium} = 0.23 - 0.08 = 0.15mol$

$n_{total} = 0.06 + 0.15 + 0.04 = 0.25mol$

$P_i = \frac{n_i}{n_{total}} \times P_{total}$

$P_{Co} = \frac{0.06}{0.25}\times 5 = 1.2bar$

$P_{H_2} = \frac{0.15}{0.25}\times 5 = 3bar$

$P_{CH_3OH} = \frac{0.04}{0.25}\times 5 = 0.8bar$

$K_p = \frac{P_{CH_3OH}}{P_{Co}\times P_{H_2}^2} = \frac{0.8}{1.2\times 3^2} = \frac{0.8}{1.2\times 9}$

$= 0.07407 = 74.07\times 10^{-3}$

74. $2Ag^+ + 2e^- \rightarrow 2Ag$

$QH_2 \rightarrow Q + 2H^+ + 2e^-$

$2Ag^+ + QH_2 \rightarrow Q + 2Ag + 2H^+$

$K = \frac{\left[H^+\right]^2}{\left[Ag^{+2}\right]^2}$

$E_{Cell} = E^0_{Cell} - \frac{0.06}{2}\log\frac{\left[H^+\right]^2}{1^2}$

$$\Rightarrow 0.4 = 0.1 - \frac{0.06}{2}\log\left[H^+\right]^2$$

$$\Rightarrow 0.3 = \frac{0.06}{2} \times 2\log\left[H^+\right]$$

$$\Rightarrow 5 = -\log\left[H^+\right]$$

$$-\log\left[H^+\right] = \frac{1}{2}\log kw + \frac{1}{2}\log\left(-\frac{1}{2}\log kb\right)$$

$$pH = 5 = \frac{1}{2} \times 14 + \frac{1}{2} \times \log\left(10^{-2}\right) - \frac{1}{2}\log kb$$

$$\Rightarrow 5 = 7 + 1 - \frac{1}{2}\log kb$$

$$= 3 = \frac{1}{2}\left(-\log kb\right)$$

$$6 = pkb$$

75. Cobalt has highest value of $E^0_{Co^{+3}/Co^{+2}}$

The complex may be $\left[Co\left(CN\right)_6\right]^{-4}$

CN is a strong field ligand.

The configuration $t_{2g}^6 e_g^1$

No. of electrons in e_g is 1